

How the Blockchain Can Help a City Get Smart



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As the digital economy grows and matures, the smart cities movement is gaining momentum. For many people this represents the promise of high-tech cities, with autonomous cars on the streets, drones delivering food, and connected devices helping city dwellers to perform a myriad of activities. This article investigates the role blockchain can play in the administration of a smart city.

The 21st century is the century of cities. There is a clear trend of population aggregation in medium-sized and big cities, with urban areas growing all over the world and competing for economic investment, a skilled workforce and cultural prominence. Some cities leading this trend, such as Shanghai, Singapore, Tokyo, London, Dubai or New York, among many others, are already world business and technology hubs, with their world significance (and GDP) sometimes outstripping that of many countries.

However, this increasing importance of cities in the world's economy does

come with a price. Population growth, environmental demands and economic downturns bring many challenges to cities all over the world, and not only to the largest ones.

Gassman, Böhm and Palmié have summarised the most important challenges cities face today: new urbanisation trends that reshape cities and their surroundings, changes in lifestyles and preferences, demographic developments (like the increasing number of elderly people in western countries), overloaded infrastructure, poor air quality, smog, noise, mobility issues, tourism, limited resources and management of the urban commons.

What is a smart city?

The definition of a smart city given above is broad. However, there are a few things that smart cities have in common. The most important one is the *digital shadow*. It is a prerequisite for any smart city to create a raw input of data about the city, which will come from an interconnected network of sensors, devices, and other digital services and products. Just imagine a city where thousands (or even hundreds of thousands) of sensors record information about air quality, temperature, traffic, energy consumption in buildings, smog, garbage production, or humidity. The